

Bangji Yang

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EDUCATION

University of Illinois Urbana-Champaign (UIUC)

Aug 2025 – Present

Master of Science in Computer Science (MSCS)

Urbana, IL

– GPA: 4.0/4.0

– Coursework: Computer Vision, Software Engineering, Machine Learning, Bioinformatics.

Tsinghua University

Aug 2021 – June 2025

Bachelor of Engineering in Automation

Beijing, China

– GPA: 3.75/4.00

– Awards: Comprehensive Excellence Award, Sports Performance Award.

– Coursework: Automatic Control, Operations Research, Random Mathematics and Statistics, Discrete Mathematics.

SELECTED PUBLICATIONS & PREPRINTS

Batched Contextual Reinforcement: A Task-Scaling Law for Efficient Reasoning

Bangji Yang, Hongbo Ma, Jiajun Fan, Ge Liu .

ICML 2026 (arXiv:2604.02322).

M3: High-fidelity Text-to-Image Generation via Multi-Modal, Multi-Agent and Multi-Round Visual Reasoning

Bangji Yang, Ruihan Guo, Jiajun Fan, Chaoran Cheng, Ge Liu.

Preprint (arXiv:2602.06166).

MutAtlas: A PDB-Wide Energy-Guided Atlas of Protein Mutation Effects

Ruihan Guo, Chaoran Cheng, Zhanghan Ni, Neil He, **Bangji Yang**, Ge Liu

ICML 2026

ConfRover: Simultaneous Modeling of Protein Conformation and Dynamics via Autoregression

Yuning Shen, Lihao Wang, Huizhuo Yuan, Yan Wang, **Bangji Yang**, Quanquan Gu.

NeurIPS 2025.

INDUSTRY EXPERIENCE

ByteDance Seed (AI Drug Design)

June 2024 – Aug 2025

Research Intern (Advisor: Prof. Quanquan Gu)

Shanghai, China

– Developed ConfRover, an autoregressive generative framework unifying SE(3) Diffusion and causal transformers to simultaneously model protein conformation and dynamics.

– Optimized large-scale inference and unified trajectory simulation, systematically outperforming traditional molecular dynamics in recovering dynamic modes.

ACADEMIC RESEARCH

Siebel School of Computing & Data Science, UIUC

Aug 2025 – Present

Graduate Research Assistant (Advisor: Prof. Ge Liu)

Urbana, IL

– **Efficient LLM Reasoning:** Pioneered **Batched Contextual Reinforcement (BCR)**, an SFT-free GRPO paradigm that enforces an implicit information bottleneck via multi-problem batching. Formulated a novel *task-scaling law* across 0.6B–8B models, demonstrating that increased parameter capacity uniquely sustains accuracy at higher batch sizes (N) while exponentially compressing reasoning trajectories.

– **Reasoning Optimization Dynamics:** Deconstructing the RL optimization manifold to bypass its inefficient stochastic exploration. Utilizing CKA analysis to trace homologous evolution in distilled models, we perform rigorous *singular value extraction* on LoRA trajectories to isolate deterministic pathways for direct, $O(1)$ reasoning optimization.

– **Multi-Agent Systems:** Designed **M3**, a closed-loop Vision-Language Model framework featuring inference-time self-refinement to dynamically correct spatial and physical errors in text-to-image generation.

– **Masked Diffusion LMs:** Enhancing model generation efficiency and stability through optimized unmasking strategies, novel sampling frameworks, and reinforcement learning integration.

Tsinghua University

Mar 2022 – June 2024

Undergraduate Researcher

Beijing, China

TECHNICAL SKILLS

Programming Languages: C/C++/C#, Python, Verilog, LaTeX

Machine Learning: PyTorch, Transformers (Hugging Face), Diffusion Models, Reinforcement Learning (PPO, GRPO), Large Language Models (LLM), Multi-Agent Systems, Generative Models

Tools & Platforms: Git, Docker, Linux/Bash, WandB